Aim of Therapy of DKA

1. Correct dehydration
2. Correct acidosis and reverse ketosis
3. Restore Blood Glucose (BG) near normal
4. Avoid complication of treatment
5. Identify and treat the precipitating event

Definition of DKA

Hyperglycemia: BG > 11mmol/l
Hyperketonemia or hyperketonuria
Metabolic acidosis, venous blood gas: PH < 7.3 +/or Bicarbonate < 15 mmol/l

Severity of DKA

Degree of acidosis	PH	Bicarbonate
Mild	< 7.3	<15mmol/l
Moderate	<7.2	<10mmol/l
Severe	<7.1	<5mmol/l

Precipitating Factors:

Missed or Inadequate insulin doses
Intercurrent illness; infection or stress
Surgery or trauma

Symptoms and Signs:

Polyuria
Polydipsia
Nocturnal enuresis
Dehydration
Rapid deep breathing
Abdominal pain
Acetone smell of breath
Disturbed level of consciousness

Investigations:

Serum	Urine
Random blood glucose	Glucose
Urea and Electrolytes	Ketone
Blood gas	Culture and sensitivity
Bone profile	
CBC	
osmolality	
culture	

Monitoring:
Clinical:

- Keep NPO, with 24 hours intake/output chart
- Cardiac monitoring
- Hourly Heart Rate(HR),Respiratory Rate (RR), Blood Pressure (BP), level of consciousness.
- Hourly neurological assessment for warning sing and symptoms of cerebral edema.
- 4 hourly temperature checking
- Nasogastric tube (NGT) for unconscious or vomiting patient.

Biochemical:

- Hourly BG by reflow check
- Urea and electrolytes on presentation, then after 2 hours then 4-6 hourly as indicated till patient is stable.
- 2 hourly venous blood gas
- Urine, whenever passed, to be checked for ketones.

Management

Emergency Assessment:

1. ABC
2. Clinical evaluation to confirm the diagnosis and carefully look for source of infection.
3. Weigh the patient, the weight should be used for calculation and not weight from previous visit or hospital record.

4. Assess clinical severity of dehydration. It should be 5-10% for usual DKA presentation. However, if > 10% dehydration is suggested, it should be justified by the presence of weak or impalpable peripheral pulse, hypotension or oliguria.
5. In rare case of shock, the circulatory volume should rapidly be restored with isotonic saline in 20ml per kg bolus.
6. Assess level of consciousness.

Indications for PICU Admission:

1. **Young age <= 2 years**
2. **Severe presentation**
3. **Disturb level of consciousness at presentation or during management.**
4. **Severe electrolytes abnormalities (e.g.: hypernatremia)**
5. **Suspected cerebral edema at presentation or during management.**

Fluid Therapy:

1. Based on clinical assessment of 5-10% dehydration, start with 10ml/kg of normal saline over 1-2 hours and may be repeated if necessary.
2. Subsequent fluid management (maintenance over 24 hours plus deficit over 36-48 hours) should be with an isotonic solution (0.9% saline) for at least 4–6 hours.
3. After 6 hours fluid can be changed to half normal saline (0.45% NACL).
4. The total fluid per day should not exceed 1.5-2 times the usual daily maintenance requirement based on weight.
5. The patient should be assessed frequently for the correction of dehydration and the fluid is adjusted accordingly.
6. Monitor BG hourly, when it fall < 15mmol/l change the fluid to 0.45%NS + 5% Dextrose.

Note:
If serum osmolality is > 360 mosmol/l, then the deficit should be given over 48 hours.

Insulin Therapy:

1. Regular insulin IV infusion to be started 1 hour after starting fluid replacement therapy, i.e. after the patient has received initial volume expansion.
2. Infusion rate: 0.05–0.1unit/kg/h {e.g., one method is to dilute 50 units regular (soluble) insulin in 50mL normal saline, 1 unit=1mL}.
3. Accept drop of glucose up to 8mmol/l in the first hour and no more than 5-6mmol/l thereafter.
4. If BG drops fast and acidosis persist, change the IV Dextrose concentration to 10% or if needed 15% depend on blood glucose.
5. If acidosis is corrected and BG > 15mmol/l persist, reduce the rate of insulin to 0.05u/kg/hr.
6. When the acidosis is corrected and patient tolerates oral feed, subcutaneous (SC) Insulin R may be started. SC insulin is started 30 minutes before a meal and IV insulin stopped 30 minutes after the SC insulin.

   Subcutaneous insulin dose: 1 unit/kg/day
   - i. 2/7 before breakfast
   - ii. 2/7 before lunch
   - iii. 2/7 before dinner
   - iv. 1/7 bed time

Note: If any newly diagnosed diabetes is admitted with hyperglycemia and ketosis BUT no acidosis, hydrate with 0.45NS at maintenance rate with 10 mmolKcl/ 500cc. continue oral feed and start SC insulin as per previous steps.

Correction of Electrolyte Disturbance

1. Potassium (K) supplement of 40mmol/L should be added to the IVF if there is no concerns about renal failure and patient passed urine in the preceding 4 hours.
2. Adjust K when serum electrolytes available:
   - i. If K dropped < 3.5mmol/l, increase supplement to 60mmol/l.
   - ii. If K > 5mmol/l, reduce supplement to 20mmol/l
3. K supplement may be given as 50% KCL and 50% KPO4 in case of severe hypophosphatemia. Monitor serum calcium and Phosphate in such cases.

Note:

- Hyperglycemia cause a dilutional effect, depressing serum sodium concentration:
- Corrected sodium = sodium (mmol/l) + Glucose (mmol/l) – 5.6/ 5.6x 1.6
- If corrected sodium > 150mmol/l, hypernatremia state exist and fluid therapy should be given over 48 hours.

Correction of Acid Base Imbalance:

- Acidosis gets corrected spontaneously with the correction of dehydration. Routine bicarbonate (HCO3) administration in case of DKA is not indicated and can lead to complications.
- Indication of HCO3 if:
    - PH <6.9 after initial resuscitation
    - Patient having shock
    - Life-threatening hyperkalemia and arrhythmias
- Dose of bicarbonate: 1-2mmol/kg over 2 hours.

Management of Cerebral edema in DKA

Background

Cerebral edema in diabetic ketoacidosis (CEDKA) is a serious complication and the incidence of clinically overt cerebral edema is 0.5–0.9% and the mortality rate is 21–24%. Mental status abnormalities (GCS scores <14), however, occur in approximately 15% of children treated for DKA and are associated with evidence of cerebral edema on neuroimaging.

Risk factors:

- Younger age
- New onset diabetes
- Longer duration of symptoms

Potential risk factors at diagnosis or during treatment of DKA.

- Greater hypocapnia at presentation after adjusting for degree of acidosis
- Increased serum urea nitrogen at presentation
- More severe acidosis at presentation
- Bicarbonate treatment for correction of acidosis
- An attenuated rise in serum sodium concentration
- Greater volumes of fluid given in the first 4h
- Administration of insulin in the first hour of fluid treatment

Signs and symptoms of cerebral edema include:

- Headache and slowing of heart rate
- Change in neurological status (restlessness, irritability, increased drowsiness, and incontinence)
- Specific neurological signs (e.g., cranial nerve palsies, papilledema)
- Rising blood pressure
- Decreased O2 saturation

Treatment

Initiate treatment as soon as the condition is suspected.

- Reduce the rate of fluid administration by one-third.
- Give mannitol, 0.5–1g/kg IV over 10–15min, and repeat if there is no initial response in 30min to 2h
- Hypertonic saline (3%), suggested dose 2.5–5mL/kg over 10–15min, may be used as an alternative to mannitol, especially if there is no initial response to mannitol.
- Hyperosmolar agents should be readily available at the bedside.
- Elevate the head of the bed to 30°.
- Intubation may be necessary for the patient with impending respiratory failure.
- After treatment for cerebral edema has been started, cranial imaging may be considered as with any critically ill patient with encephalopathy or acute focal neurologic deficit. The primary concern is whether the patient has a lesion requiring emergency neurosurgery (e.g., intracranial hemorrhage) or a lesion that may necessitate anticoagulation (e.g., cerebrovascular thrombosis).

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
ISPAD Clinical Practice Consensus Guidelines		2014	
Pediatric Diabetes	Wolfsdorf JI	2014	154–179

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